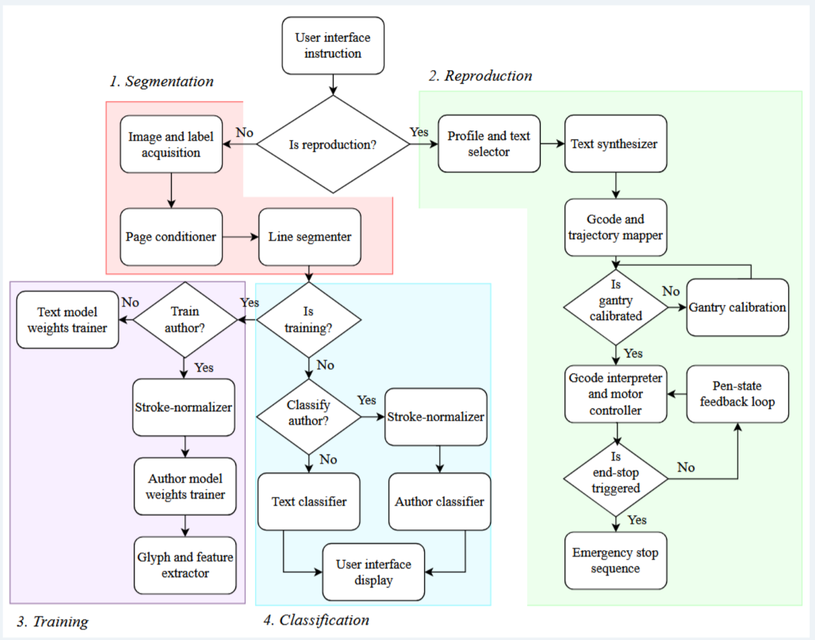
Handwriting Robot

Dylan Braune

Supervisor: Prof H.C. Myburgh

Control Systems Group
Department of Electrical, Electronic and Computer Engineering



Project overview

The system reads photographed handwriting and identifies the writer. It uses stored glyph shapes to reproduce new text in a selected style with a pen gantry. The system supports 11 writer styles.

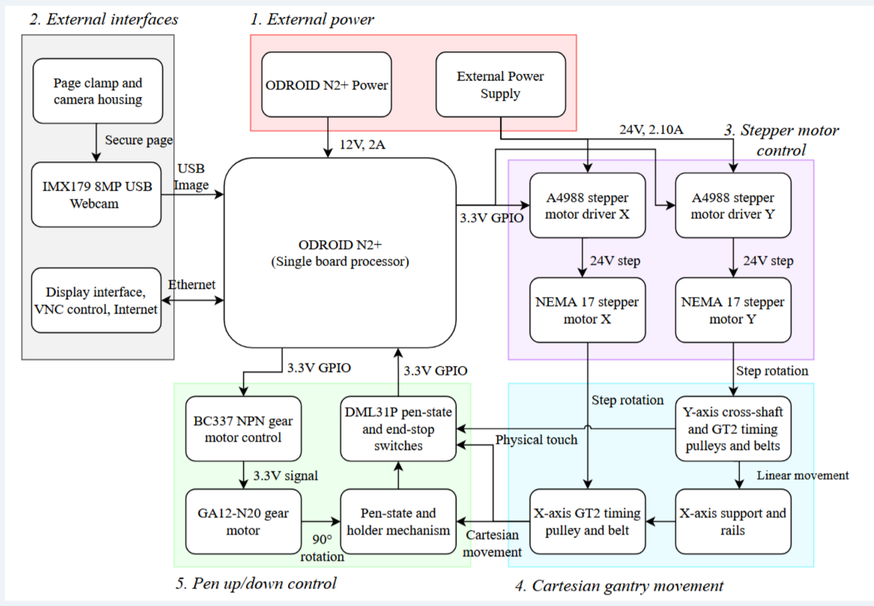
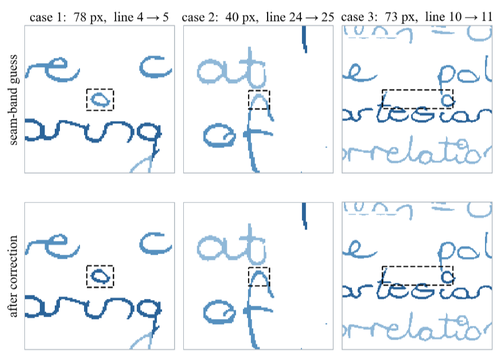
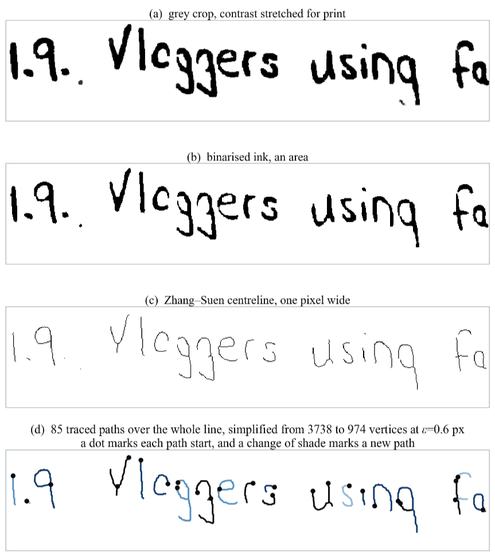


Image processing

The system corrects uneven lighting and page rotation. It separates the writing into lines and traces the ink into stroke paths.

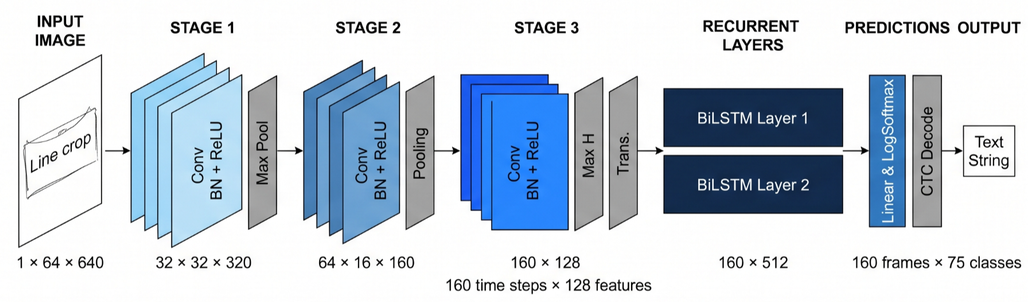




Recognition

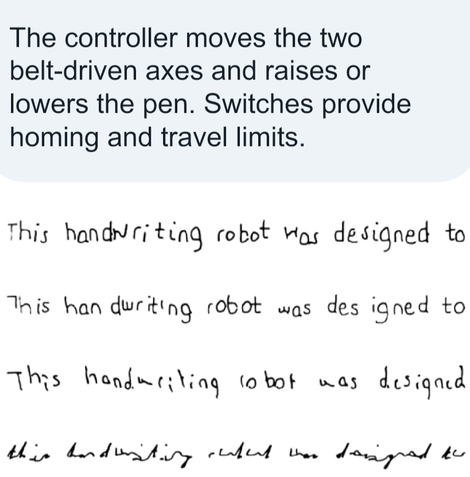
A neural network reads the text in each line. A second network identifies the writer after stroke width is normalised.

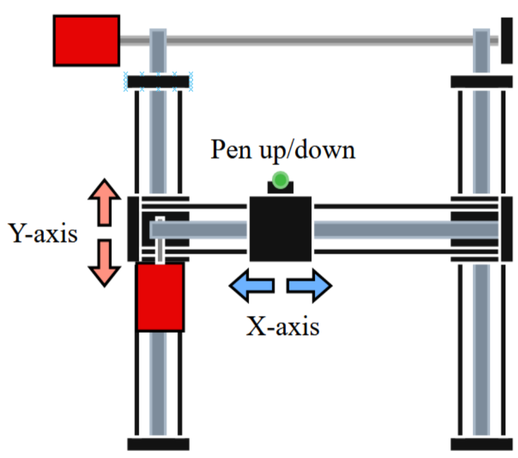
Parameter	Setting
Reader input	64 × 640 px
Writer styles	11
Stroke width	0.17 x-height



Motion control

The controller moves the two belt-driven axes and raises or lowers the pen. Switches provide homing and travel limits.

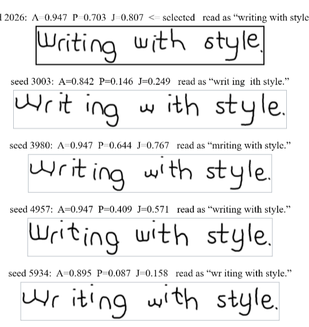


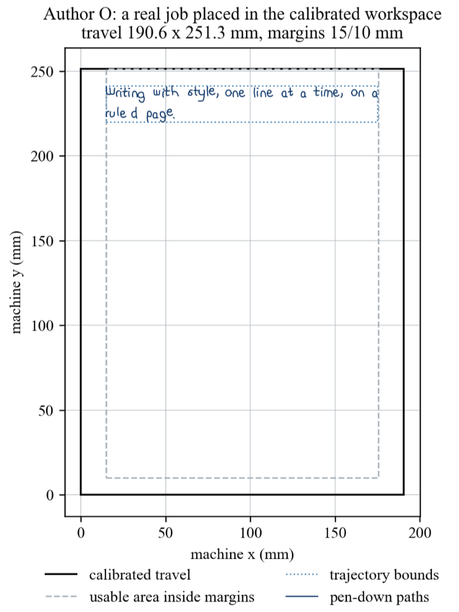


Handwriting synthesis

The system selects stored glyphs and sets their spacing and joins. It converts the resulting paths into G-code for the gantry.







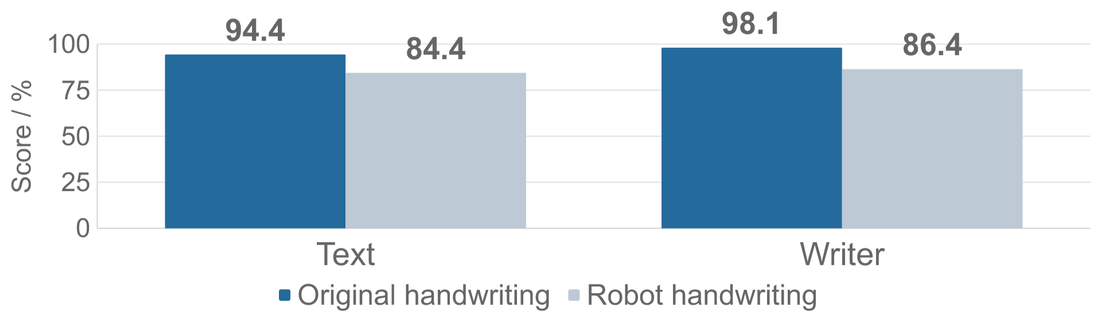
Measured performance

86.7%
relative shape score
against natural handwriting variation

0.297 mm
mean absolute length error

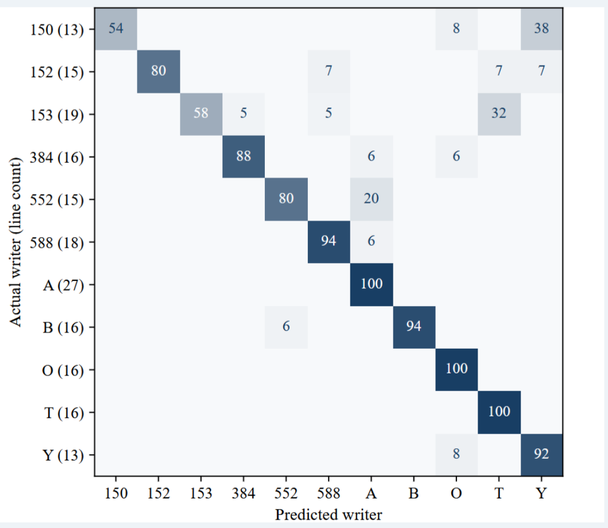
2.43 s
average completion time per character, 25 jobs

Recognition scores



Original validation and robot output are separate test sets.
Classification time remains above target.

Robot writer identification



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie / Lefapha la Boetšenere, Tikologo ya Kago le Theknolotši ya Tshedimošo